

**RecursiveMinimumEigenOptimizer with QAOA throws ValueError.**

<https://github.com/qiskit-community/qiskit-optimization/issues/264>

ROW 103

# RecursiveMinimumEigenOptimizer with QAOA throws ValueError. #264

[New issue](#)[Closed](#)

lasys opened on Oct 5, 2021

...

## Information

- Qiskit Optimization version: 0.2.2
- Qiskit version: 0.29.0
- Python version: 3.8.6
- Operating system: Linux

## What is the current behavior?

RecursiveMinimumEigenOptimizer / pauli\_trotter\_evolution throws following error:

```
-----
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_2491547/4017197336.py in <module>
     31
     32 # execute recursive qaoa
--> 33 RecursiveMinimumEigenOptimizer(MinimumEigenOptimizer(qaoa)).solve(max_cut_qubo)

~/local/lib/python3.8/site-packages/qiskit_optimization/algorithms/recursive_minimum_eigen_optimizer.py in solve(self, problem, max_iter)
    231
    232     # solve current problem with optimizer
--> 233     res = self._optimizer.solve(problem_) # type: OptimizationResult
    234     if self._history == IntermediateResult.ALL_ITERATIONS:
    235         optimization_results.append(res)

~/local/lib/python3.8/site-packages/qiskit_optimization/algorithms/minimum_eigen_optimizer.py in solve(self, operator, offset, problem_)
    195     operator, offset = problem_.to_ising()
    196
--> 197     return self._solve_internal(operator, offset, problem_, problem)
    198
    199     def _solve_internal(

~/local/lib/python3.8/site-packages/qiskit_optimization/algorithms/minimum_eigen_optimizer.py in _solve_internal(self, operator, offset, problem_, problem)
    208     if operator.num_qubits > 0:
    209         # approximate ground state of operator using min eigen solver
--> 210         eigen_result = self._min_eigen_solver.compute_minimum_eigenvalue(operator)
    211         # analyze results
    212         raw_samples = None

~/local/lib/python3.8/site-packages/qiskit/algorithms/minimum_eigen_solvers/vqe.py in compute_minimum_eigenvalue(self, operator, numshots)
    449     # this sets the size of the ansatz, so it must be called before the initial point
    450     # validation
--> 451     self._check_operator_ansatz(operator)
    452
    453     # set an expectation for this algorithm run (will be reset to None at the end)

~/local/lib/python3.8/site-packages/qiskit/algorithms/minimum_eigen_solvers/qaoa.py in _check_operator_ansatz(self, operator)
    129     # Recreates a circuit based on operator parameter.
    130     if operator.num_qubits != self.ansatz.num_qubits:
--> 131         self.ansatz = QAOAAnsatz(
    132             operator, self._reps, initial_state=self._initial_state, mixer_operator=self._mixer
    133         ).decompose() # TODO remove decompose once #6674 is fixed

~/local/lib/python3.8/site-packages/qiskit/circuit/quantumcircuit.py in decompose(self)
   1290
   1291     pass_ = Decompose()
-> 1292     decomposed_dag = pass_.run(circuit_to_dag(self))
   1293     return dag_to_circuit(decomposed_dag)
   1294

~/local/lib/python3.8/site-packages/qiskit/converters/circuit_to_dag.py in circuit_to_dag(circuit)
    53     dagcircuit.add_clbits(circuit.clbits)
    54
--> 55     for register in circuit.qregs:
    56         dagcircuit.add_qreg(register)
    57

~/local/lib/python3.8/site-packages/qiskit/circuit/library/evolved_operator_ansatz.py in qregs(self)
    151     """A list of the quantum registers associated with the circuit."""
    152     if self._data is None:
--> 153         self._build()
    154     return self._qregs
    155
```

## Assignees

No one assigned

## Labels

No labels

## Type

No type

## Projects

No projects

## Milestone

No milestone

## Relationships

None yet

## Development

No branches or pull requests

## Notifications

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## Participants



```

~/local/lib/python3.8/site-packages/qiskit/circuit/library/evolved_operator_ansatz.py in _build(self)
    202         continue
    203
--> 204         evolved_op = self.evolution.convert((coeff * op).exp_i()).reduce()
    205         circuits.append(evolved_op.to_circuit())
    206         is_evolved_operator.append(True) # has time coeff

~/local/lib/python3.8/site-packages/qiskit/opflow/evolutions/pauli_trotter_evolution.py in convert(self, op)
    101         # Sort into commuting groups
    102         operator = self._grouper.convert(operator).reduce()
--> 103         return self._recursive_convert(operator)
    104
    105     def _recursive_convert(self, operator: OperatorBase) -> OperatorBase:

~/local/lib/python3.8/site-packages/qiskit/opflow/evolutions/pauli_trotter_evolution.py in _recursive_convert(self, operator)
    145         return circuit_no_identities
    146     elif isinstance(operator.primitive, PauliOp):
--> 147         return self.evolution_for_pauli(operator.primitive)
    148     # Covers ListOp, ComposedOp, TensoredOp
    149     elif isinstance(operator.primitive, ListOp):

~/local/lib/python3.8/site-packages/qiskit/opflow/evolutions/pauli_trotter_evolution.py in evolution_for_pauli(self, pauli_op)
    175     # to produce correct CoB circuit
    176     sig_bits = np.logical_or(pauli_op.primitive.z, pauli_op.primitive.x)
--> 177     a_sig_bit = int(max(np.extract(sig_bits, np.arange(pauli_op.num_qubits)[::-1])))
    178     destination = (I.tensorpower(a_sig_bit)) ^ (Z * pauli_op.coeff)
    179     cob = PauliBasisChange(destination_basis=destination, replacement_fn=replacement_fn)

ValueError: max() arg is an empty sequence

```

## Steps to reproduce the problem

```

import networkx as nx
import numpy as np
from qiskit.algorithms.optimizers import SPSA
from qiskit.utils import QuantumInstance
from qiskit_optimization.algorithms import MinimumEigenOptimizer, RecursiveMinimumEigenOptimizer

from qiskit.algorithms import QAOA
from qiskit_optimization.applications import Maxcut
from qiskit import Aer
from qiskit_optimization.problems.quadratic_program import QuadraticProgram
from qiskit_optimization.converters import QuadraticProgramToQubo

# create graph
n = 4
graph = nx.Graph()
graph.add_nodes_from(np.arange(0, n, 1))
elist = [(0, 1, 1), (0, 3, 1), (0, 2, 1), (1, 2, 1), (1, 3, 1), (2, 3, 1)]
graph.add_weighted_edges_from(elist)

# create Maxcut
max_cut = Maxcut(graph)
max_cut_problem = max_cut.to_quadratic_program()
conv_toQubo = QuadraticProgramToQubo()
max_cut_qubo = conv_toQubo.convert(max_cut_problem)

# create qaoa
reps = 3
initial_point = [-0.54695145, 3.42723337] * reps
quantum_instance = QuantumInstance(backend=Aer.get_backend('qasm_simulator'), shots=1024)
qaoa = QAOA(optimizer=SPSA(maxiter=0), reps=reps, initial_point=initial_point, quantum_instance=quantum_instance)

# execute recursive qaoa
RecursiveMinimumEigenOptimizer(MinimumEigenOptimizer(qaoa)).solve(max_cut_qubo)

```

## What is the expected behavior?

The program runs without errors.

## Suggested solutions

None.

It seems to be a sporadic error that does not always occur.

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I have tested the function using the same scheme with several graphs and I noticed that the error happens mainly with unweighted graphs.



t-imamichi on Nov 4, 2021 · edited by t-imamichi

Edits ▾ Collaborator ⋮

`RecursiveMinimumEigenOptimizer` iteratively solves the problem by fixing some variables. The subproblems have smaller number of variables than that of the original problem. So, setting `initial_point` explicitly does not work because the size of `initial_point` can be larger than that of subproblems.

This PR fixes the case where a subproblem is an identity. [Qiskit/qiskit#7225](https://github.com/Qiskit/qiskit/pull/7225)



t-imamichi on Nov 9, 2021

Collaborator ⋮

If you still have an issue, please reopen this.



 t-imamichi closed this as **completed** on Nov 9, 2021